# Myanmar Earthquake & Dam Risk Report

## 1. Executive Summary

This report covers 9,242 earthquake events in Myanmar from 1970-07-29 to 2026-04-14. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 0.350 with magnitude of completeness Mc=2.4.

**Total Events:** 9,242

**Date Range:** 1970-07-29 to 2026-04-14

**Max Magnitude:** M7.7

**Mean Depth:** 24.5 km

**b-value:** 0.350

**a-value:** 4.685

**Mc (completeness):** 2.4

## 2. Earthquake Map

## 3. Seismic Clustering Analysis

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## 4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

**Critical Risk:** 25 dams

**High Risk:** 113 dams

**Moderate Risk:** 9 dams

**Low Risk:** 107 dams

**Top 30 Highest Risk Dams:**

| Name | River | PGA(g) | Dist Fault | Risk | Grade |
|---|---|---|---|---|---|
| Dam near 22.09N 95.86E | | 0.3715 | 5.7 km | 8.3 | Critical |
| Dam near 21.77N 95.89E | | 0.4680 | 2.7 km | 8.3 | Critical |
| Natkan | | 0.3190 | 7.9 km | 8.2 | Critical |
| Dam near 21.47N 95.87E | | 0.3650 | 6.0 km | 8.2 | Critical |
| Ta Nai Hka | Ta Nai Hka | 0.4288 | 3.9 km | 8.2 | Critical |
| Dam near 20.72N 95.90E | | 0.3756 | 5.6 km | 8.2 | Critical |
| Dam near 21.74N 95.85E | | 0.3298 | 7.4 km | 8.2 | Critical |
| Taungnyo | Taungnyo | 0.5083 | 1.2 km | 8.2 | Critical |
| Dam near 20.60N 95.90E | | 0.3293 | 7.4 km | 8.2 | Critical |
| Dam near 20.47N 96.08E | | 0.2806 | 9.9 km | 8.2 | Critical |
| Dam near 20.85N 96.03E | | 0.3097 | 8.3 km | 8.2 | Critical |
| Meikhtila | | 0.3113 | 8.2 km | 8.2 | Critical |
| Dam near 17.55N 96.37E | | 0.2866 | 9.6 km | 8.1 | Critical |
| Dam near 21.05N 95.83E | | 0.2639 | 11.0 km | 8.0 | Critical |
| Dam near 21.04N 95.83E | | 0.2658 | 10.9 km | 8.0 | Critical |
| Thaphan Chaung | Thaphan Chaung | 0.2566 | 11.5 km | 7.8 | Critical |
| Dam near 21.34N 95.81E | | 0.2476 | 12.2 km | 7.7 | Critical |
| Kun Chaung | Kun Chaung | 0.5128 | 0.9 km | 7.6 | Critical |
| Phyu Chaung | Pyu Chaung | 0.5064 | 1.3 km | 7.5 | Critical |
| Kabaung | Kabaung Chaung | 0.2920 | 9.2 km | 7.5 | Critical |
| Dam near 21.37N 95.80E | | 0.2340 | 13.4 km | 7.5 | Critical |
| Yenwe | Ye Nwe Chaung | 0.4291 | 3.9 km | 7.4 | Critical |
| Nyaunggone | | 0.2301 | 13.7 km | 7.4 | Critical |
| Dam near 20.73N 95.83E | | 0.2318 | 13.6 km | 7.4 | Critical |
| Zeedaw | Nyaung Oat | 0.2142 | 15.3 km | 7.2 | Critical |
| Male-Nattaung | Male-Nattaung | 0.1883 | 18.6 km | 6.9 | High |
| Kyee Ni | | 0.1835 | 19.4 km | 6.8 | High |
| Kintat | Mu | 0.1737 | 21.0 km | 6.7 | High |
| Dam near 20.58N 96.17E | | 0.1755 | 20.7 km | 6.7 | High |
| Myotha | Ku Htaw | 0.1762 | 20.6 km | 6.7 | High |

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## 5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

**K (productivity):** 0.96

**c (time offset):** 0.119 hours

**p (decay exponent):** 0.831

**Expected M>=3 (7d):** 0

**Expected M>=3 (30d):** 0

**Expected M>=3 (90d):** 0

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

## 6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Boore & Joyner (1997) attenuation model
- Report generated by mmeq-opendata toolkit